ExpertRL-HPO

Meta-feature-aware RL meets Budget-smart BOHB for Lightweight AutoML

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duoleveling_plus1

Modality 3/3

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Literature

Motivation

Context:

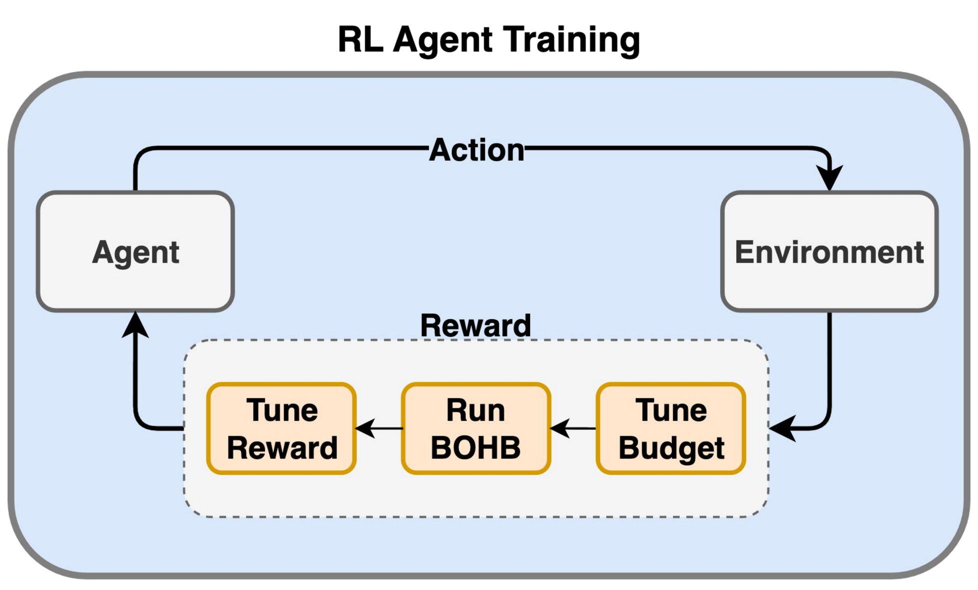
- **Dynamic Algorithm Configuration (DAC)** with deep RL can *adapt* model complexity to each dataset, but training an RL agent demands thousands of costly pipeline evaluations.
- **Multi-fidelity Bayesian optimisation (BOHB)** cuts search cost by evaluating many hyper-parameter sets on tiny data slices, yet it assumes the *pipeline structure* is fixed in advance.
- For large-scale text-classification challenges (heterogeneous datasets, tight GPU budgets, pressure for quick turnaround) neither approach alone is ideal, each needs an external signal to tune the process.

Proposed Approach:

We aim to solve this by combining **Meta-feature** aware **RL-based DAC** with **BOHB** in a *single loop*, letting the RL agent pick models and the optimizer fine-tune them on a budget.

ExpertRL-HPO

ExpertRL-HPO couples meta-feature-aware RL with multi-fidelity BOHB to deliver a cost-efficient, interpretable, *one-click* AutoML pipeline for text-classification.

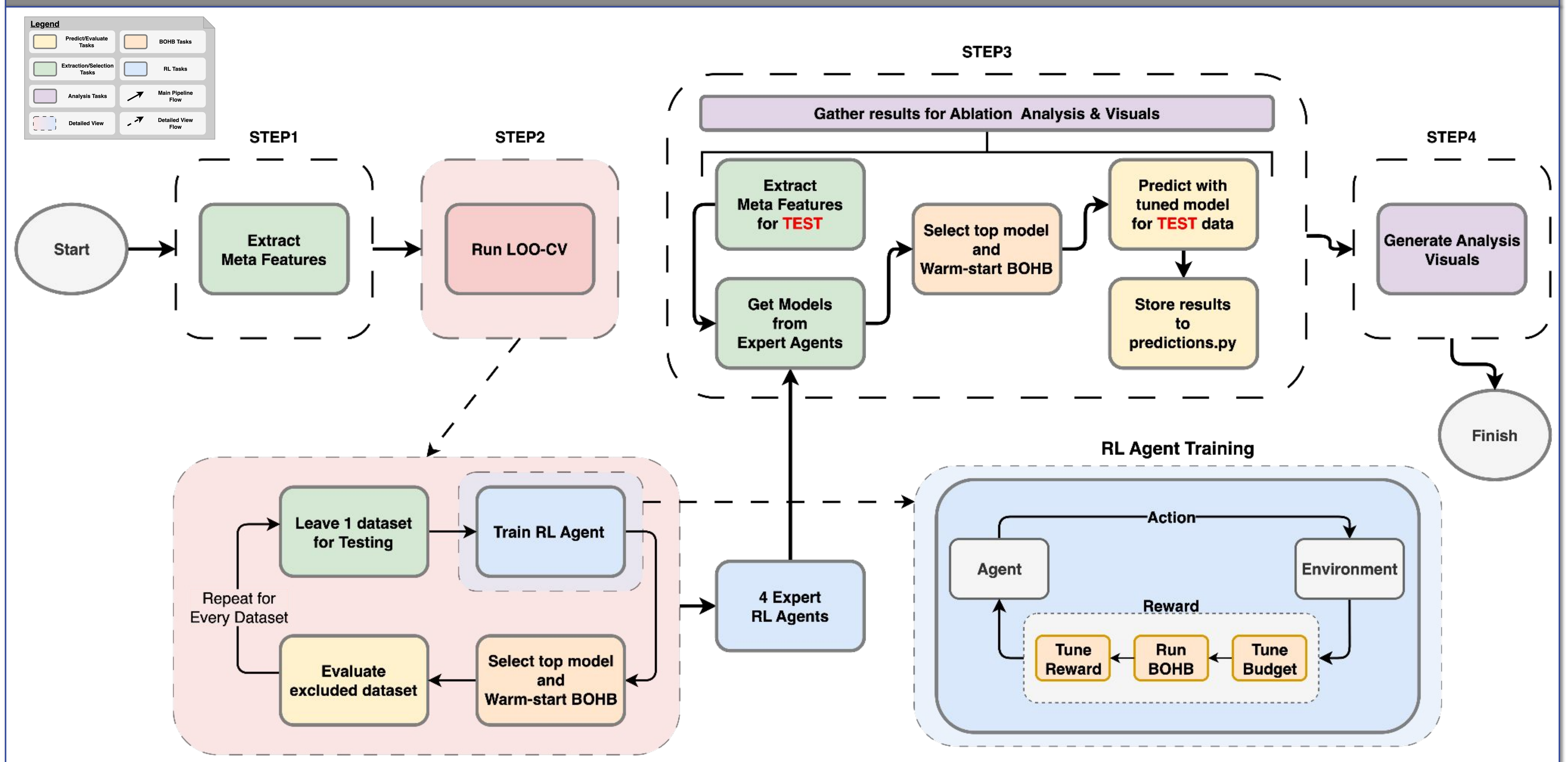


MDP formulation:

- **State (s):**
  - 37-dim meta-feature vector.
- **Action (a)**: select **model tier**
  - simple: (TF-IDF+LR / SVM)
  - medium: (LSA+LR)
  - complex: (TF-IDF+MLP)
- **Transition Function:**
  - $P(S_{t+1} = s \mid s_t, a_t) = 1 \quad \forall a_t$
  - next state = current state
- **Reward r(s,a):**
  - 0.5 x accuracy
    − 0.2 x complexity
    + 0.15 x confidence-gap
    + 0.15 x richness

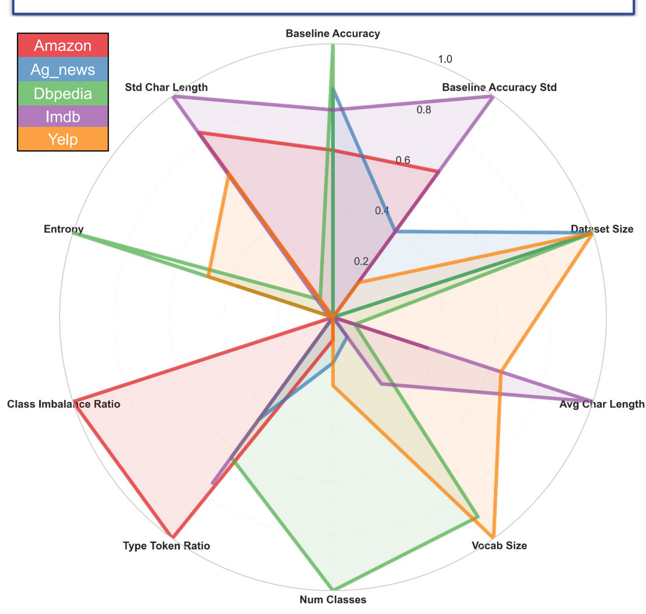
Our Approach

Full AutoML Pipeline



Results

How Meta-Features Guide the RL Agents



LOO-CV Agent Results

Table 1: LOO-CV Agent Results

Dataset	Model	Acc. (%)
Amazon	medium	81.0
AG News	medium	86.0
DBPedia	medium	92.0
IMDb	medium	82.0

Best Yelp Result

Table 2: Best Yelp Result

Dataset	Model	Acc. (%)
Yelp	complex	**63.6**

Resources Used

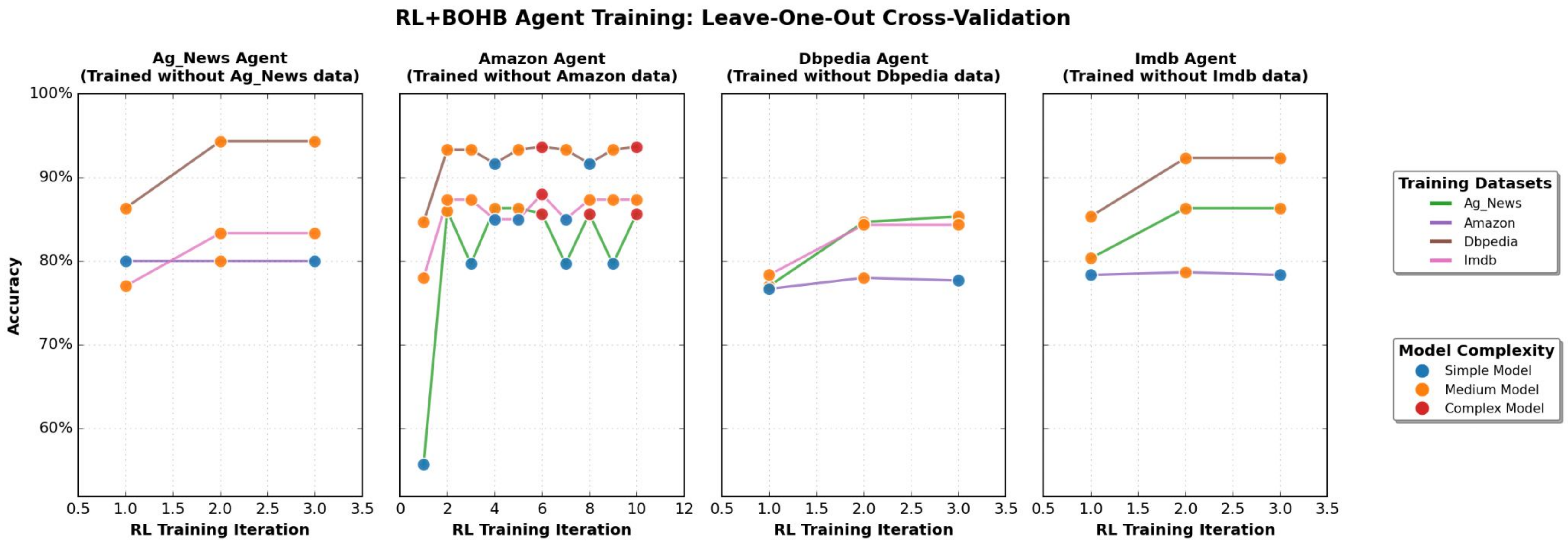
For development:
- 1 RTX2080 GPU
- 1 Intel Core i7 CPU
- 1 Intel Core i5 CPU
- Total compute estimate: 600 CPU-h

For AutoML:
- Same sources
- 8-10 h

Workforce:
- 2 full week on average

Number of queries for test score generation: **3**

Analysis of Pipeline Run



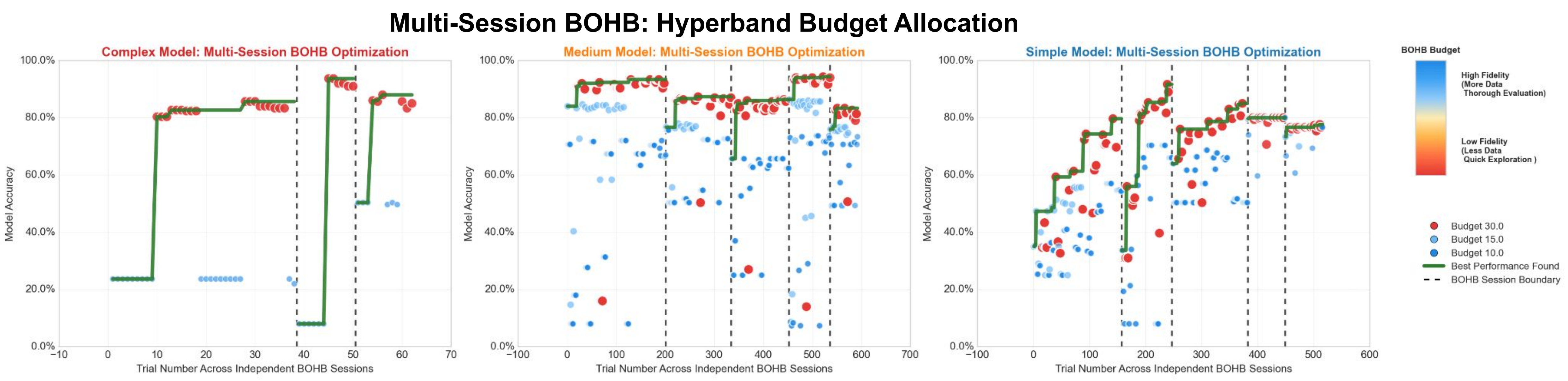


Table 3: CV-Agents Performances (Model (Accuracy%))

Amazon Agent	AG_News Agent	DBPedia Agent	IMDb Agent
medium (55.6%)	medium (55.6%)	medium (55.6%)	medium (55.6%)

Limitations

- **Compute-Heavy Models Omitted:** Large transformer backbones were skipped due to hardware limits.
- **Limited Training Scope**: The RL agent was not trained on four datasets; broader data could improve generalisation.

Future Work

- **Broaden Model Benchmarks**: Evaluate the pipeline with additional SOTA architectures (BERT, RoBERTa, Longformer).
- **Reward Shaping**: Feed BOHB-measured accuracy directly into the RL reward to tighten the optimisation loop.
- **Meta-Feature Engineering**: Refine and weight meta-features based on their learned contribution to rewards and agent decisions.